

Motivation and Related Work (draft)

PINN + Mamba Digital Twin for Mecanum Forward Dynamics and Friction Identification

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1 Motivation

Four-wheel Mecanum platforms are attractive across industrial and service settings—warehouse and intralogistics transport, manufacturing and assembly feeding, hospital and clinical logistics, cleaning, inspection, and assistive mobility—because their omnidirectional maneuverability lets them translate and rotate independently in cluttered, human-shared, space-constrained environments where conventional differential-drive vehicles require multi-point turns [1]. Yet a recent systematic review of Mecanum and omni-wheel technologies identifies four recurring barriers that gate real-world deployment: limited stability and maneuverability on uneven and inclined terrain, high energy consumption, complex control requirements, and elevated cost [1].

1.1 Why higher-fidelity control is needed

Underneath those application-level barriers sits a common set of physics-level obstacles that the ideal-rolling kinematic model omits, and each has been shown to cause a concrete, independently reported failure. *Energy*: in a real industrial haul—five tonnes of zinc ingots transported 5 km in an iron-steel plant—wheel slippage caused $\approx 6.18\%$ energy loss, cutting battery-supported range proportionally, which a kinematic controller assuming ideal rolling cannot account for [2]. *Accuracy*: roller-surface contact produces significant deviations in vehicle orientation during tracking, requiring a control-layer intervention (a ramp-shaped reference) beyond baseline tracking to suppress [3]; and the resulting odometric drift is surface- and velocity-dependent, so the same kinematic controller yields different positioning error across floors and speeds [4]. *Coupled accuracy/energy*: restricted force/torque authority causes transient overshoot that simultaneously lowers tracking performance and raises energy use, reducing autonomy [5]. *Root cause*: Mecanum friction is comparatively undermodeled relative to pneumatic or railroad wheels, and an experimentally validated orthotropic-friction model shows the contact behavior depends on both velocity and the payload-shifted center of mass [6]. Because these effects are nonlinear, load-dependent, and surface-dependent, purely kinematic or partially dynamic controllers cannot reject them—precisely why higher-fidelity model-based control and system identification are required [7, 8].

Most directly for this work, on the *KUKA youBot* platform itself the real Mecanum-wheel construction (a finite number of rollers and their arrangement) degrades odometric navigation whenever the odometry algorithm treats the wheel as an idealized disc, and an algorithm that accounts for the true geometry measurably reduces the coordinate error [9]. This on-platform evidence motivates the friction/dynamics fidelity pursued here (the simulator’s LuGre–Adamov friction model follows the same line of youBot-specific modeling).

1.2 Sector barriers and where control is decisive

The same physics surfaces as distinct barriers across deployment sectors. In *heavy industry*, slip-driven energy loss shrinks effective range on heavy-load hauls [2]. In *automotive* intralogistics, baseline controllers lack the robustness for a four-Mecanum autonomous vehicle, motivating adaptive sliding-mode control with an extended state observer [10]. In *warehouse* logistics, surface/velocity-dependent odometry drift undermines the pose repeatability that automated workflows depend on [4]. In *aerospace* assembly, tolerances on the order of 0.01 mm coexist with mobile robots whose position-control accuracy and structural stiffness are low, so deployment remains largely at the proof-of-concept stage and relies on two-stage positioning—coarse by the platform wheels, fine by a laser-tracker-guided arm [11]. In *hospitals*, mobile service robots must satisfy six risk dimensions spanning environmental complexity, hygiene, human interaction, workflow flexibility, and autonomy [12], and their navigation testing is still preliminary and not standardized [13].

A recurring pattern ties these together. The commercial precision bar is set by platforms such as KUKA omniMove, which moves loads up to 90 tonnes and states contactless positioning to ± 3 mm [14]; yet its base drive accuracy is on the order of ± 5 mm and is tightened to $\approx \pm 1$ mm only with add-on optical tracking, positioning aids, and mechanical guides [15]—the same external-metrology workaround seen in aerospace, where docking accuracy and repeatability are treated as a first-class requirement [16]. In other words, deployed precision is bought by external referencing rather than by higher-fidelity onboard control, and Mecanum slip/odometry drift is exactly what forces that dependence [4, 9]. The barriers thus split cleanly: *control-related*—slip-induced orientation error, odometric drift, disturbance/uncertainty rejection, and docking repeatability—versus *non-control*—cost, battery life, systems integration, standards compliance, hygiene, and workforce. This paper targets the former.

1.3 Deployability near humans favors interpretable, verifiable models

A further, deployment-blocking barrier is regulatory rather than merely performance-related. Driverless industrial trucks—the category covering Mecanum AGVs and AMRs—are governed by ISO 3691-4:2023, which mandates safety functions rated to Performance Levels (typically PL d, corresponding to SIL 2), personnel detection, and speed-limited operating zones; it inherits its method from ISO 13849-1 and IEC 61508, and human-proximity collaboration adds ISO 10218 and ISO/TS 15066 [17, 18]. Against this backdrop, systematic reviews find that neural-network (black-box) controllers are confined to the lowest safety- integrity levels precisely because their outputs cannot be exhaustively predicted or formally verified, with interpretability a named open problem [19, 20]. This directly favors a *physics-informed*, structurally-constrained model—one that retains a known friction structure and identifies interpretable per-wheel parameters—over an unconstrained black box, both because it is more amenable to the verification and PL rating these standards demand and because the resulting twin can itself serve as a verification and reachability tool for the deployed controller.

The core difficulty underlying all of the above is friction. Slip and stick-slip behavior at the roller contact are governed by dynamic friction that simplified constant- or Coulomb-plus-viscous models do not capture, and whose parameters are neither directly measurable nor constant across surfaces and payloads. A digital twin that both predicts forward dynamics at high fidelity and *recovers* the underlying per-wheel friction parameters therefore addresses the root cause rather than compensating for its symptoms downstream—the objective of this work.

Gap addressed	Approach	Reported gain
Model fidelity $\rightarrow$ tracking error	Complete Lagrangian + linearized motor delay [7]	DTW 0.2662 m vs. 0.3198 m ($\sim 17\%$)
Disturbance rejection (load, noise)	IMC + Disturbance Observer [7]	97.6% / 98.3% faster settling vs. IMC / PID
Inclined surfaces (slip, gravity)	ADRC [21]	49–60% better tracking vs. PD (experimental)
Unknown CoG offset + slope	MRAC, Lyapunov adaptation [22]	Real-time compensation; stability proof
Disturbance + model uncertainty	AVPSMO feedforward in MPC [8]	Lyapunov-proven suppression

Table 1: Higher-fidelity model-based control on Mecanum platforms. All figures are the cited authors’ own results against their own baselines.

2 Related Work

2.1 Higher-fidelity model-based control closes measurable gaps

A consistent body of recent Mecanum-specific results shows that improving model fidelity and adding explicit disturbance estimation measurably improves tracking and robustness (Table 1). A complete Lagrangian model that linearizes the PWM-induced motor delay reduces open-loop Dynamic-Time-Warping (DTW) trajectory error to 0.2662 m versus 0.3198 m for a traditional model ($\sim 17\%$ improvement) [7]. Coupling Internal Model Control with a Disturbance Observer (IMC+DOB) stabilizes 97.6% faster than IMC and 98.3% faster than PID under load and noise disturbance [7]. Active Disturbance Rejection Control (ADRC), which estimates and compensates unknown dynamics with minimal plant knowledge, improves inclined-surface tracking accuracy by 49–60% over a PD baseline with experimental validation [21]. Adaptive and observer-based schemes go further by estimating the disturbing quantities themselves: a Model Reference Adaptive Control law with Lyapunov-based adaptation compensates unknown CoG offset and slope in real time [22], and an adaptive variable-power sliding-mode observer used as a feedforward compensator within an MPC (AVPSMO-MPC) suppresses external disturbances and model uncertainty with a Lyapunov-proven stability guarantee [8]. The common thread is that the residual physics—slip, friction, and load—must be modeled or estimated, not ignored; all reported gains are the respective authors’ own results against their own baselines.

2.2 Deployed control standard versus the research frontier

There is a wide gap between what industry ships and what the literature proposes. Commercial Mecanum platforms are driven, in practice, by kinematic inverse-Mecanum allocation with per-wheel PID beneath a SLAM/navigation stack, and their positioning accuracy is secured largely through mechanical precision and external referencing rather than advanced control—the KUKA omniMove specification of ± 3 mm (base ± 5 mm, ± 1 mm only with optical/mechanical aids) is representative [14, 15], and youBot-class navigation likewise rests on kinematic odometry [9]. The research frontier, by contrast, is dominated by robust and adaptive designs that repeatedly demonstrate the nominal/kinematic controller to be insufficient under real disturbance: adaptive integral terminal sliding mode [23], anti-peaking linear ADRC [5], sliding-mode-observer MPC [8], and adaptive sliding mode with an extended state observer for automotive intralogistics [10].

Yet none of these is standard deployed practice, and—crucially for this work—none *recovers the per-wheel friction parameters online*. This is the gap the proposed PINN + Mamba digital twin targets: closing the fidelity/identifiability distance between deployed kinematic control and the disturbance/slip/friction reality the frontier has documented.

2.3 Physics-informed learning for friction and forward dynamics

Physics-informed neural networks (PINNs) are an established route to recovering friction parameters and learning forward dynamics while preserving physical consistency. Most directly relevant, a PINN framework fuses the LuGre dynamic friction model with learnable neural components to identify LuGre parameters from small, noisy datasets at higher fidelity than simplified friction models, validated on an underactuated nonlinear system [24]—the same identification problem this paper poses for per-wheel (μ, χ) recovery. More broadly, embedding robot dynamics into the learner improves both identification and prediction: a PINN that encodes joint dynamics inside a recurrent network with customized Runge–Kutta cells outperforms conventional PINNs and grey-box state-space identification and matches ground-truth parameters on UR3e hardware [25], and enhanced PINNs that additionally model motor dynamics and external-force coupling report 31.12%/37.07% modeling-accuracy gains and 40.31%/51.79% joint-tracking-error reductions over prior state-of-the-art [26]. For the nonsmooth behavior central to slip, PINNs that embed nonsmooth multi-body formulations capture stick–slip and separation–reattachment without the vanishingly small time steps of conventional time-stepping [27], and Kolmogorov–Arnold-network methods recover interpretable closed-form static-friction laws ($R^2 > 0.95$) with no predefined functional form [28]. Notably, rigid-body-prior networks alone are insufficient: standard Deep Lagrangian Networks fail to capture nonlinear friction and flexibility, motivating augmentations such as DeLaN-FFNN that add a learnable non-conservative component [29]—direct justification for pairing a physics prior with a learnable friction term rather than relying on either alone.

2.4 Selective state-space models as a sequence backbone

Trajectory data for a richly-excited Mecanum platform are long, and the friction state (e.g. LuGre bristle deflection) is history-dependent, motivating a sequence model that scales. Mamba introduces selective state-space models with input-dependent state transitions, achieving linear-time complexity in sequence length and demonstrated improvements on real data up to million-length sequences [30], providing an efficient long-horizon backbone for the PINN + Mamba digital twin proposed here.

2.5 The gap this paper addresses

Two observations define the opening. First, the strongest PINN friction-ID and forward-dynamics evidence [25, 26, 28, 29] is drawn from fixed-base industrial manipulators and cobot joints, not from Mecanum omnidirectional mobile robots; the single most on-point friction-ID result [24] is validated on one underactuated testbed. Their transfer to *per-wheel* friction (μ, χ) recovery on a four-Mecanum platform is thus established only by analogy. Second, to our knowledge no prior work applies a selective state-space (Mamba) backbone to Mecanum or mobile-robot forward-dynamics learning. This paper connects these threads: a physics-informed, Mamba-backed digital twin that learns forward dynamics and recovers interpretable per-wheel friction parameters for a KUKA youBot Mecanum platform, targeting exactly the slip/friction/CoG chokepoint that Section 2.1 shows to be the binding constraint on Mecanum control fidelity.

Level	PFH _d (dangerous failures/h)	≈SIL	What it permits for an AMR near people
none / PL a	$\geq 10^{-5}$	—	Not safety-creditable; the function may run only <i>behind</i> an independent certified safety layer, or in segregated/caged space with no humans present.
PL b	$3 \times 10^{-6} - 10^{-5}$	SIL 1	Low-consequence safety functions (warnings, non-critical interlocks).
PL c	$10^{-6} - 3 \times 10^{-6}$	SIL 1	Moderate functions, e.g. reverse / low-speed travel where risk is lower.
PL d	$10^{-7} - 10^{-6}$	SIL 2	<i>Workhorse level for shared workspaces</i> : personnel detection, protective stop, speed-and-separation monitoring, forward travel among humans.
PL e	$10^{-8} - 10^{-7}$	SIL 3	Highest-hazard cases (high mass/speed, severe-injury potential); rarely required for standard AMRs.

Table 2: ISO 13849-1 Performance Levels, their target dangerous-failure rates and approximate IEC 61508 SIL mapping, and what each permits for a Mecanum AMR operating near humans [17, 18, 31].

A Certifiability Implications of Interpretable, Physics-Informed Control

Deploying an AMR in close proximity to humans is gated less by nominal tracking accuracy than by *functional-safety certification*. Under ISO 3691-4:2023, driverless industrial trucks realize their protective functions—personnel detection, protective stop, and speed-and-separation monitoring—through the methodology of ISO 13849-1, whose Performance Levels (PL a–e) map onto the Safety Integrity Levels of IEC 61508 [17, 18, 31]. Operation in a shared human workspace typically requires the relevant safety function to reach **PL d (SIL 2)** in the forward direction (Table 2). The decisive obstacle for learning-based control is *systematic capability*: these standards demand evidence that the control software is verifiable, output-bounded, structurally testable, and free of systematic faults. A black-box neural controller cannot furnish this evidence—its outputs cannot be exhaustively bounded, its structure cannot be covered by test, and its behavior cannot be argued deterministic—which is why systematic reviews find neural controllers “confined to the lowest levels of safety integrity” [19, 20].

This is precisely where the physics-informed structure of the proposed twin changes the achievable certification *status*—though not, by itself, the Performance Level. A model that retains a known friction structure (LuGre) and rigid-body constraints, identifies interpretable per-wheel parameters (μ, χ) with physical bounds, and admits provable output bounds and reachability supplies exactly the systematic-capability and verification evidence that a black box cannot [24, 28]. The effect is a change of role rather than an automatic level jump: the learned component moves from *non-creditable* (effectively SIL 0, requiring it to be fully fenced off or overridden) to *verifiable and creditable* within the safety argument (plausibly SIL 1–2 for its role). Reaching PL d/SIL 2 still requires the full ISO 13849 package—Category-3 architecture, diagnostic coverage, and hardware fault tolerance—so interpretability unlocks the software/systematic half of the case, not the whole

of it. In practical terms it yields two deployment gains: (i) a verifiable, bounded forward-dynamics model lets the performance channel operate less conservatively (higher speed, tighter separation) at the *same* certified PL, reducing reliance on external metrology and nuisance stops; and (ii) the physics-grounded twin can serve as a certifiable runtime monitor—its divergence from measured motion is a diagnosable, bounded trip condition—and as the reachable-set oracle a safety assessor requires. We emphasize that certification is ultimately an assessor’s safety-case judgment; interpretability is an *enabler of the required evidence*, not a certificate in itself [19, 20].

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